

Program : Diploma in Automation and Robotics	
Course Code : 5332	Course Title: AI for Robotics
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To Understand the importance, applicability, and strength of AI.
- To Describe the learning methods adopted in AI
- To understand the architecture and types of Expert system.
- To Interpret various applications of AI in Robotics

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize students with Artificial Intelligence principles and techniques	14	Understanding
CO2	Understand learning concepts and learning from outcomes	15	Applying
CO3	Understand the importance, applicability, and strength of AI.	15	Applying
CO4	Familiarize with programming languages in AI and understand the architecture of Expert system.	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1					
CO2	2	1					
CO3	2	1					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize students with Artificial Intelligence principles and techniques		
M1.01	Familiarize with basic AI concepts and historical background	4	Understanding
M1.02	Understand various application areas of AI	6	Understanding
M1.03	Understand about the basics of expert systems	4	Understanding
Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview			
CO2	Understand learning concepts and learning from outcomes		
M2.01	Understand the basic types of learning	6	Understanding
M2.02	Discuss the details of reinforcement learning	3	Understanding
M2.03	Familiarize with programming languages used in AI	6	Applying
	Series Test – I	1	
Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG.			
CO3	Understand the importance, applicability, and strength of AI.		
M3.01	Discuss different features and organization of expert systems	5	Understanding
M3.02	Understand the knowledge representation schemes in expert systems	4	Understanding
M3.03	Familiarize with expert system tools	6	
EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN.			

CO4	Familiarize with programming languages in AI and understand the architecture of Expert system.		
M4.01	Familiarize with AI applications pertaining to robotics	10	
M4.02	Discuss the ethics and risks of AI usage in robotics	4	
	Series Test – II	1	
AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics			

Text / Reference

T/R	Book Title/Author
T1	Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson Education, India,2016
T2	Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business Media.
R1	Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) 2013
R2	N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, 2009.
R3	Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, Harlow:AddisonWesley,2002